

OVN Job Search Proof of Concept

Technical Stack Brief | Date: April 14, 2026

Executive Summary: The OVN Job Search application is a frontend-only React + TypeScript proof of concept built with Vite. It generates compliant outbound job search links to external platforms (no scraping, no automation), with strong input validation and a modular structure designed for rapid iteration and low operational overhead.

1. Architecture at a Glance

- **Application Type:** Single-page web application (SPA)
- **Architecture:** Client-only (no backend service in v1)
- **Core Workflow:** User enters ZIP, radius, role, and keywords; app validates input; app generates outbound search URLs for supported job platforms.
- **Compliance Position:** URL generation only; users are redirected to original job boards in new tabs.

2. Core Technology Stack

Layer	Technology	Version / Notes	Purpose
Language	TypeScript	v5.6.x	Type safety, maintainability, predictable refactoring.
UI Framework	React	v18.3.x	Component-based UI for form input and generated results panel.
DOM Renderer	react-dom	v18.3.x	Browser rendering for React components.
Build Tool / Dev Server	Vite	v5.4.x	Fast local development, optimized production bundling.
React Build Integration	@vitejs/plugin-react	v4.3.x	JSX transform and React tooling within Vite.

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Type Definitions	@types/react, @types/react-dom, @types/node	18.x / 22.x	Compile-time typing support for framework and runtime APIs.

3. Delivery and Runtime Model

- **Local Development:** npm run dev (Vite dev server)
- **Production Build:** npm run build (TypeScript project build + Vite bundle)
- **Preview Build:** npm run preview
- **Deployment Posture:** Static asset output suitable for static hosting; repository currently configured with a Vite base path for GitHub Pages deployment pattern.

4. Functional Scope Implemented

Inputs and Validation

- ZIP code (5 digits)
- Radius in miles (validated numeric range)
- Required role/title
- Up to three optional keywords

Search Variants

- Base role search
- Role + Keywords search
- Single Keyword search only when exactly one keyword is provided

Current Platform Integrations

- LinkedIn
- Indeed
- The Ladders
- ZipRecruiter
- Health eCareers
- PracticeLink
- ASCO Career Center (US-only, radial location search format)

5. Compliance and Risk Controls (v1)

- No scraping of third-party sites.
- No automated login to third-party platforms.
- No browser automation against partner job boards.

- Outbound URL generation only, preserving platform terms-friendly behavior.

6. Codebase Design Quality

- **Separation of Concerns:** URL builders, validation logic, label generation, and search result generation are isolated in utility modules.
- **UI Modularity:** Form and results are split into reusable React components.
- **Extensibility:** Adding a new platform is straightforward by adding one URL builder and one entry in result generation flow.
- **Maintainability:** Strong TypeScript typing for search input, validation errors, platforms, and result variants.

7. Recommended Next Technical Steps

1. Add automated unit tests for URL builders and validation logic.
2. Add lightweight analytics for click-through tracking per platform and variant.
3. Add environment-based configuration for platform enable/disable and future feature flags.
4. Evaluate optional backend service only if persistence, personalization, or reporting becomes a requirement.

Prepared from the current repository implementation and project conventions as of April 14, 2026.